AMERICAN ACADEMY OF OPHTHALMOLOGY®

Double-Masked, Dose-Response, Vehicle-Controlled Study of VVN461 Ophthalmic Solution in Postoperative Ocular Inflammation

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Purpose: To evaluate the safety and ocular efficacy of VVN461, a Janus kinase kinase inhibitor, for treatment of postoperative inflammation.

Design: This was a phase II, multicenter, double-masked, randomized, vehicle-controlled, parallel-comparison study in subjects who underwent routine unilateral cataract extraction and lens replacement surgery via phacoemulsification without surgical complication.

Participants: Ninety-one subjects at 9 private practice US ophthalmology offices.

Intervention: VVN61 0.5% and 1.0% topical ophthalmic solution or vehicle, 4 times daily.

Main Outcome Measures: The proportion of subjects with anterior chamber cell (ACC) grade 0 in the study eye at day 14.

Results: The proportion of subjects with ACC grade 0 in the study eye at day 14 was 60.0% (18/30), 53.3% (16/30), and 19.4% (6/31) in the VVN461, 1.0%, VVN461, 0.5%, and vehicle groups, respectively ($P = 0.0012$ and 0.0057). Treatment effects in favor of VVN461 were also seen in the need for rescue medication and reduction of anterior chamber flare. There were relatively few adverse events in either VVN461 treatment group (10%, 3/30, or less), and they were judged of mild severity.

Conclusions: In this double-masked, vehicle-controlled study, clinically and statistically significant anti-inflammatory efficacy was seen with few safety issues.

Financial Disclosure(s): Proprietary or commercial disclosure may be found in the Footnotes and Disclosures at the end of this article. *Ophthalmology Science 2025;5:100806* *© 2025 by the American Academy of Ophthalmology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).*

Supplemental material available at www.ophthalmologyscience.org.

Intraocular inflammation is an anticipated sequela of intraocular surgery, such as cataract extraction and lens replacement surgery, and is manifested principally as conjunctival injection, corneal edema, ciliary flush, and aqueous cells and flare. In general, trauma to the internal structures of the eye is accompanied by the production of prostaglandins and other vasoactive molecules, the release of proinflammatory cytokines, an increase in blood flow to the affected area, and extravasation of protein and cellular blood elements. Untreated inflammation may lead to complications, such as cystoid macular edema and corneal scarring. Thus, managing and treating postoperative inflammation is an important goal after cataract surgery.[1]

Current postoperative medication regimens commonly include topical corticosteroids. Treatment with corticosteroids is employed to reduce pain and discomfort and to facilitate recovery of the blood-aqueous barrier. When administered at the time of surgery and during the immediate postoperative period, corticosteroids can reverse the clinical manifestations of inflammation. In the United States, topical corticosteroids are routinely prescribed with up to 4 times daily dosing for ≥2 weeks after cataract surgery.[1]

The Janus kinase (JAK) pathway plays a key role in inflammatory cell regulation, cytokine production, and proinflammatory signal transduction. Dysregulation of the JAK pathway is associated with the pathogenesis of various inflammatory and autoimmune disorders. Therefore, JAK inhibitors have the potential to alleviate the inflammatory process. The JAK-signal transducer and activator of transcription pathway is known to be important for inflammatory cell regulation, cytokine production, and proinflammatory signal transduction.[2]

VVN461 is a newly synthesized molecule. It is a potent inhibitor of JAK1 ($IC_{50} = 0.082$ nm), tyrosine kinase 2 ($IC_{50} = 0.55$ nM), a slightly less potent inhibitor of JAK2 ($IC_{50} = 1.36$ nM), and a weak inhibitor of JAK3 ($IC_{50} = 15.92$ nM). Nonclinical ocular and systemic safety studies were conducted to support an Investigational New Drug application with the US Food and Drug Administration (Data on File, VivaVision Pharmaceuticals).

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https://doi.org/10.1016/j.xops.2025.100806
ISSN 2666-9145/25

Methods

Study Design

This was a phase II, multicenter, double-masked, randomized, vehicle-controlled, parallel-comparison study assessing the safety and ocular efficacy of VVN461 for treating postoperative ocular inflammation in subjects who underwent routine unilateral cataract extraction and lens replacement surgery via phacoemulsification without surgical complication. Approximately 90 completed subjects (30 per group) were planned to be randomized in a 1:1:1 ratio. Subjects administered 1 eye drop in the study eye four times daily for 14 days. The investigational product (IP) was sterile isotonic solutions of VVN461, 1.0%; VVN461, 0.5%; or vehicle, preserved with benzalkonium chloride and packed in identical eye dropper bottles. VVN461, 1.0% had a pH of 5.0; VVN461, 0.5% had a pH of 4.9; and the vehicle had a pH of 4.9, which met the a priori established specification of 4.5 to 6.5. There were a total of 7 visits over approximately 8 weeks for each subject. This study was conducted at 10 private practice sites (Table S1, available at www.ophthalmologyscience.org) in the United States under an Investigational New Drug exemption in accordance with Good Clinical Practice as required by the US Food and Drug Administration regulations. The study was approved by an institutional review board, adhered to the Declaration of Helsinki, and all subjects provided written informed consent before enrollment in the study. This study was registered on clinicaltrials.gov as NCT06164743.

Anterior chamber inflammatory cells were graded as 0 (0) to 4+ (>50 cells per field), and flare was graded as 0 (none) to 4+ (intense) according to the Standardization of Uveitis Nomenclature scale.[3] Pain was measured using a 0 (no pain) to 10 (worst pain imaginable) Numeric Pain Rating Scale. Adverse events (AEs) were coded using the Medical Dictionary for Regulatory Activities (MedDRA, version 24.1) system.

Subject Eligibility

Eligible for inclusion were individuals ≥21 years of age who were planning to undergo routine unilateral cataract extraction and lens replacement surgery via phacoemulsification extraction and implantation of an intraocular lens. At visit 1 (screening), individuals had to have the potential, in the opinion of the investigator, for postoperative best-corrected visual acuity ≤0.2 logarithm of the minimum angle of resolution (i.e., 20/32 Snellen or better) in the study eye as assessed using ETDRS and best-corrected visual acuity ≤1.0 logarithm of the minimum angle of resolution (i.e., 20/200 Snellen) in the nonstudy eye due to pathology other than cataract as assessed using ETDRS. Individuals also had to demonstrate their ability to self-administer eyedrops.[4] Excluded were individuals with ocular pain, recent or planned use of corticosteroids, oral or topical nonsteroidal anti-inflammatory drugs, or Omidria (i.e., phenylephrine and ketorolac injection), recent incisional ocular surgery, or clinically significant ocular pathology at screening or intraocular pressure (IOP) ≥24 mmHg at visit 1 (screening). Contact lens use was prohibited during the study. Women of childbearing potential were excluded if they were pregnant, nursing, or planning a pregnancy during the study and were required to use an acceptable method of contraception throughout the study.

Statistics

A priori, the primary population for analyses was the full analysis set, which consisted of all subjects who were randomized. The primary estimand was treatment difference between VVN461 (1.0% or 0.5%) and the vehicle in the proportion of subjects with anterior chamber cell (ACC) grade 0 at visit 6 (day 14) in the study eye using the full analysis set. Pearson chi-square was used to test the primary endpoint between the 2 different doses of VVN461 versus vehicle. Analyses were also performed on observed data without failure criteria for rescue. Computations for all results were performed using SAS (version 9.4, SAS/STAT 15.2) computer software package (SAS Institute, Inc, 2013, 2020). Statistical significance was defined as $P \leq 0.05$, 2-tailed. For the primary efficacy measure, testing was conducted in hierarchical order as H1 (high concentration of active vs. vehicle) and H2 (low concentration of active vs. vehicle). If H1 was not significant, then H2 will not be tested. Additional efficacy analyses performed are presented as nominal P values.

A priori, with a sample size in each group of 30, the study had 80% power to detect a difference of 35%, assuming a response rate of 30% in the vehicle group and a 2-sided 0.05 significance. P values are reported as 2 digits to the right of the decimal point.

Results

Disposition, Demographics, and Baseline Characteristics

The investigators screened 127 subjects, of whom 91 were randomized to treatment. All subjects in all groups completed the study (Fig S1, available at www.ophthalmologyscience.org). While a per-protocol subset was planned, there were no major protocol violations, and so it was not required.

Demographics and Other Baseline Characteristics

The demographics and other baseline characteristics of the study population are shown in Table 1. For the most part, demographic characteristics were not appreciably different among the 3 treatment groups. The mean age of the study population was 69.2 years (range: 52 to 85 years). The majority of subjects were ≥65 years of age (66/91 [72.5%] subjects) and female (55/91 [60.4%] subjects). The most common race was White (69/91 [75.8%] subjects), followed by Black or African American (12/91 [13.2%] subjects) and Asian (5/91 [5.5%] subjects). Ethnically, approximately a quarter of subjects (25/91 [27.5%]) were Hispanic or Latino. The proportion of female subjects was higher in the VVN461, 0.5%, group (23/30 [76.7%] subjects) than in the other groups (14/30 [46.7%] subjects, VVN461, 1.0%; 18/31 [58.1%] subjects, vehicle, Table 1).

Anti-inflammatory Efficacy

Primary endpoint: The proportion of subjects with ACC grade 0 in the study eye at day 14, was 60.0% (18/30), 53.3% (16/30), and 19.4% (6/31) in the VVN461, 1.0%, VVN461, 0.5%, and vehicle groups, respectively. This difference was statistically significant between the VVN461, 1.0%, and vehicle groups ($P < 0.01$). As this comparison met the statistical hierarchy to protect the alpha, the next comparison was tested. The comparison between the VVN461, 0.5%, and vehicle groups was also statistically

Table 1. Demographic Characteristics

Characteristic	VVN461, 1.0% (N = 30)	VVN461, 0.5% (N = 30)	Vehicle (N = 31)	Overall (N = 91)
Age (yrs)				
Mean (SD)	69.1 (7.28)	68.5 (8.51)	69.8 (6.79)	69.2 (7.49)
Min; Max	54; 85	52; 84	53; 81	52; 85
Age categories, n (%)				
<65 yrs	9 (30.0)	9 (30.0)	7 (22.6)	25 (27.5)
≥65 yrs	21 (70.0)	21 (70.0)	24 (77.4)	66 (72.5)
Sex, n (%)				
Male	16 (53.3)	7 (23.3)	13 (41.9)	36 (39.6)
Female	14 (46.7)	23 (76.7)	18 (58.1)	55 (60.4)
Race, n (%)				
White	23 (76.7)	24 (80.0)	22 (71.0)	69 (75.8)
Black or African American	4 (13.3)	3 (10.0)	5 (16.1)	12 (13.2)
Asian	1 (3.3)	2 (6.7)	2 (6.5)	5 (5.5)
Other	1 (3.3)	1 (3.3)	1 (3.2)	3 (3.3)
Multiple	1 (3.3)	0 (0)	1 (3.2)	2 (2.2)
Ethnicity, n (%)				
Hispanic or Latino	8 (26.7)	9 (30.0)	8 (25.8)	25 (27.5)
Non-Hispanic or Latino	22 (73.3)	21 (70.0)	23 (74.2)	66 (72.5)

SD = standard deviation.
Percentages were based on the number of subjects in each group or overall.
Individual race categories were completed only if a single choice was provided. Otherwise, multiple choices for race were summarized under the multiple category.

significant ($P < 0.01$). Also analyzed was the proportion of subjects meeting this criterion at earlier visits (Table 2 and Fig 1). Comparisons of the proportion of subjects with ACC grade 0 were not significant between the VVN461 groups at any time point ($P \geq 0.37$).

Rescue medications: Over the course of the study, approximately one-fifth (19/91 [20.9%]) of subjects received rescue medication (topical ocular corticosteroids) for postoperative ocular inflammation in the study eye. There was a noticeable difference in the use of rescue medication for ocular inflammation in subjects between groups: 10.0% (3/30) in the VVN461, 1.0% group; 3.3% (1/30) in the VVN461, 0.5% group; and 48.4% (15/31) in the vehicle group. Both VVN461 (1.0% and 0.5%) groups demonstrated superiority over the vehicle group at visit 4 (day 3) for the proportion of subjects requiring rescue medication ($P = 0.02$). The VVN461, 0.5% group continued to demonstrate superiority over the vehicle group at day 7 ($P = 0.02$). There were no statistically significant differences between the groups at day 14 (Table 3). One subject (103-010; vehicle) received prednisolone acetate in the study eye to treat a postoperative AE of ocular inflammation but was not recorded as requiring rescue.

The effect of treatment on anterior chamber flare was similar to ACC in showing the superiority of each concentration of VVN461 to vehicle in reducing inflammation (Table S2, available at www.ophthalmologyscience.org).

Pain was graded as low in all groups at all postbaseline visits. From visit 4 (day 3) through visit 6 (day 14), mean postoperative ocular pain was between 0.00 and 0.57 in the VVN461, 1.0% group; 0.03 and 1.17 in the VVN461, 0.5% group; and 0.06 and 1.50 in the vehicle group (scale of 0−10). Using a binary analysis (yes/no), at the first

Table 2. Categorical Analysis of Subjects with Anterior Chamber Cell Grade 0 in the Study Eye by Visit (FAS)

Visit	VVN461, 1.0% (N = 30)	VVN461, 0.5% (N = 30)	Vehicle (N = 31)
Visit 4 (day 3)			
n	30	30	31
Yes, n (%)	1 (3.3)	2 (6.7)	2 (6.5)
No, n (%)	29 (96.7)	28 (93.3)	29 (93.5)
Versus vehicle			
P value	0.57	0.97	
Versus VVN461, 0.5%			
P value	0.55		
Visit 5 (day 7)			
n	30	30	31
Yes, n (%)	9 (30.0)	6 (20.0)	3 (9.7)
No, n (%)	21 (70.0)	24 (80.0)	28 (90.3)
Versus vehicle			
P value	0.05	0.26	
Versus VVN461, 0.5%			
P value	0.3711		
Visit 6 (day 14)			
n	30	30	31
Yes, n (%)	18 (60.0)	16 (53.3)	6 (19.4)
No, n (%)	12 (40.0)	14 (46.7)	25 (80.6)
Versus vehicle			
P value	<0.01	<0.01	
Versus VVN461, 0.5%			
P value	0.60		

AE = adverse event; FAS = full analysis set.
Day 14 is primary analysis.
Subjects who were rescued (lack of efficacy) or discontinued due to an AE were considered failures (i.e., "No") from the time of the rescue and/or AE through all following visits. Subjects who received anti-inflammatory medications for AEs were included in the analysis.
P values were produced from a Pearson chi-square test.

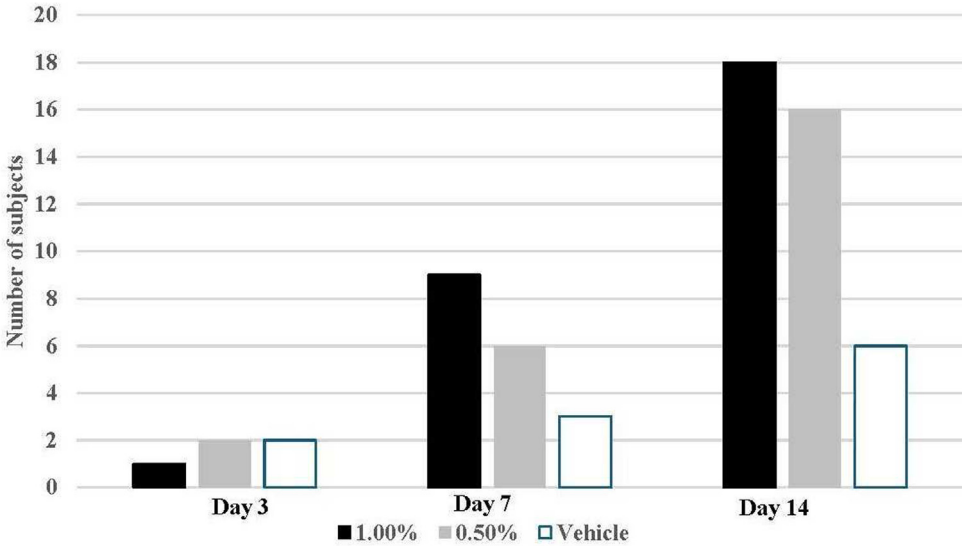


Figure 1. Number of subjects with an anterior chamber cell score of 0 by visit. Each active group was statistically significantly different from the vehicle at both day 14 and day 28 ($P < 0.01$). N = 30 to 31 for each treatment group. *P* values for comparison of VVN461 1.0% and 0.5% to vehicle: day 3: 0.57 and 0.9729; day 7: 0.05 and 0.26; day 14: <0.01 and <0.01, respectively.

postoperative visit, 58% of subjects (53/91) reported no pain. This proportion increased over the 2 weeks of the study to 77% of subjects (70/91). While the proportion of subjects in the VVN461 groups reporting no pain was numerically superior to the vehicle group, this difference was not statistically significant due to the relatively low number of subjects with pain in the vehicle group (Table 4).

Table 3. Categorical Analysis of Subjects Requiring Rescue Medication in the Study Eye by Visit (FAS)

Visit	VVN461, 1.0% (N = 30)	VVN461, 0.5% (N = 30)	Vehicle (N = 31)
Total subjects rescued, n (%)	3 (10.0)	1 (3.3)	15 (48.4)
Visit 4 (day 3)			
n	30	30	31
Yes, n (%)	0 (0)	0 (0)	5 (16.1)
No, n (%)	30 (100.0)	30 (100.0)	26 (83.9)
Versus vehicle			
P value	0.02	0.02	
Versus VVN461, 0.5%			
P value	–		
Visit 5 (day 7)			
n	30	30	31
Yes, n (%)	1 (3.3)	0 (0)	5 (16.1)
No, n (%)	29 (96.7)	30 (100.0)	26 (83.9)
Versus vehicle			
P value	0.09	0.02	
Versus VVN461, 0.5%			
P value	0.3132		
Visit 6 (day 14)			
n	30	30	31
Yes, n (%)	2 (6.7)	1 (3.3)	5 (16.1)
No, n (%)	28 (93.3)	29 (96.7)	26 (83.9)
Versus vehicle			
P value	0.25	0.09	
Versus VVN461, 0.5%			
P value	0.55		

FAS = full analysis set.
P values were produced from a Pearson chi-square test.

Safety

AEs. Overall, there were relatively few AEs, and for the VVN461 groups, they were judged to be of mild severity. Ocular treatment emergent AEs (TEAEs): Eight subjects had ≥1 ocular TEAE in the study eye, with similar proportions among groups: 2/30 (6.7%), VVN461, 1.0%; 3/30 (10.0%), VVN461, 0.5%; and 3/31 (9.7%), vehicle. The only ocular TEAE to occur in >1 study eye was increased IOP, affecting 1 (1/30 [3.3%]) subject in the VVN461, 0.5%, group and 1 (1/31 [3.2%]) subject in the vehicle group. In the former, the event was mild and considered unrelated to IP. In the latter, the event was moderate and considered unrelated to IP. The only ocular TEAE reported in the nonstudy eye was increased IOP, in the same vehicle subject who reported the condition in the study eye. The event was moderate and considered unrelated to IP (Table 5). One (1/91 [1.1%] overall; 1/31 [3.2%], vehicle) subject discontinued IP due to an ocular TEAE (moderate eye inflammation in the study eye, unrelated to the investigation product).

Nonocular TEAEs. Two subjects had ≥1 nonocular TEAE: 1 (1/30 [3.3%]) subject in the VVN461, 1.0% group and 1 (1/31 [3.2%]) subject in the vehicle group. Each nonocular TEAE was mild and considered unrelated to the IP. For the VVN461 subject, the event (headache) was deemed unrelated to the IP. For the vehicle subject, 1 event (nausea) was considered possibly related to the IP, and another event (headache) was considered unlikely related to the IP.

Table 4. Categorical Analysis of Subjects with No Postoperative Ocular Pain (NPRS) in the Study Eye by Visit (FAS)

Visit	VVN461, 1.0% (N = 30)	VVN461, 0.5% (N = 30)	Vehicle (N = 31)
Visit 4 (day 3)			
n	30	30	26
Yes, n (%)	21 (70.0)	16 (53.3)	16 (61.5)
No, n (%)	9 (30.0)	14 (46.7)	10 (38.5)
Versus vehicle			
P value	0.50	0.54	
Versus VVN461, 0.5%			
P value	0.18		
Visit 5 (day 7)			
n	29	30	21
Yes, n (%)	27 (93.1)	24 (80.0)	13 (61.9)
No, n (%)	2 (6.9)	6 (20.0)	8 (38.1)
Versus vehicle			
P value	<0.01	0.15	
Versus VVN461, 0.5%			
P value	0.14		
Visit 6 (day 14)			
n	27	29	16
Yes, n (%)	27 (100.0)	28 (96.6)	15 (93.8)
No, n (%)	0 (0)	1 (3.4)	1 (6.3)
Versus vehicle			
P value	0.19	0.66	
Versus VVN461, 0.5%			
P value	0.33		

AE = adverse event; FAS = full analysis set; NPRS = Numeric Pain Rating Scale.
Subjects who were rescued (lack of efficacy) or discontinued due to an AE were excluded from analysis from the time of the rescue and/or AE through all following visits. Subjects who received anti-inflammatory medications for AEs were included in the analysis.
P values were produced from a Pearson chi-square test.

There were no changes of note in biomicroscopy or ophthalmoscopy. With respect to IOP, baseline mean IOP was similar among groups and between eyes (16 mmHg). Mean IOP generally decreased at postbaseline visits by 1 to 2 mmHg. No subject lost vision in the VVN461 treatment groups. There were no changes of note in heart rate or blood pressure.

Discussion

In this double-masked, vehicle-controlled, parallel study of VVN461 ophthalmic solution in subjects recovering from routine unilateral cataract surgery, both VVN461, 1.0%, and VVN461, 0.5%, were effective at managing postoperative ocular inflammation and pain. The statistically and clinically significant efficacy of VVN461 relative to the concurrent vehicle group was seen in some measures as early as the first on-treatment visit, day 3, and statistically and clinically significant efficacy continued to be observed through day 14. This efficacy was seen in the proportion of subjects with 0 ACC and anterior chamber flare. The anti-inflammatory efficacy was also seen in the investigator's decision to rescue subjects with a topical corticosteroid. Of the 19 subjects rescued, 15 of 19 (78.9%) were in the vehicle group, and only 4 of 19 (21.0%) were in the active groups. With respect to subjective pain, there was a treatment effect seen in the early postoperative period. However, the differences between groups at day 14 were not statistically significant. It is noteworthy that the high rate of rescue and discontinuation from the study due to AEs in the vehicle group makes meaningful comparisons between the groups difficult.

Table 5. Ocular TEAEs by System Organ Class and Preferred Term (SAF)

System Organ Class Preferred Term	VVN461, 1.0% (N = 30) n (%)	VVN461, 0.5% (N = 30) n (%)	Vehicle (N = 31) n (%)
Subjects with any ocular TEAE, study eye	2 (6.7)	3 (10.0)	3 (9.7)
Eye disorders	2 (6.7)	1 (3.3)	1 (3.2)
Conjunctival hyperemia	1 (3.3)	0 (0)	0 (0)
Glare	1 (3.3)	0 (0)	0 (0)
Cystoid macular edema	0 (0)	1 (3.3)	0 (0)
Eye inflammation	0 (0)	0 (0)	1 (3.2)
Infections and infestations	0 (0)	0 (0)	1 (3.2)
Conjunctivitis	0 (0)	0 (0)	1 (3.2)
Injury, poisoning, and procedural complications	0 (0)	1 (3.3)	0 (0)
Corneal abrasion	0 (0)	1 (3.3)	0 (0)
Investigations	0 (0)	1 (3.3)	1 (3.2)
Intraocular pressure increased	0 (0)	1 (3.3)	1 (3.2)
Subjects with any ocular TEAE, nonstudy eye	0 (0)	0 (0)	1 (3.2)
Investigations	0 (0)	0 (0)	1 (3.2)
Intraocular pressure increased	0 (0)	0 (0)	1 (3.2)

AE = adverse event; IP = investigational product; SAF = safety analysis set; TEAE = treatment emergent adverse event.
A TEAE was defined as an AE that started on or after the date of the first dose of double-masked IP, up to and including the last dose of double-masked IP.
If a subject experienced >1 event in a given category, that subject was counted only once in that category.
Percentages were based on the number of subjects in each group.

From a safety perspective, both ocular and nonocular AEs were relatively rare in all treatment groups and most likely related to the postoperative status of the subjects.

As typical for controlled studies of topical anti-inflammatory agents, this study was vehicle (negative) controlled. Thus, we do not have concurrent positive controls. However, the magnitude of anti-inflammatory efficacy seen with VVN461 (18/30, 60.0% for VVN461 1.0%, 16/30, 53.3% for VVN461 0.5%) relative to vehicle (6/31, 19.4%) is at least as effective as that reported with topical corticosteroids or nonsteroidal anti-inflammatory drugs, including the most recently US Food and Drug Administration-approved clobetasol propionate.[5,6]

In conclusion, the efficacy and lack of safety issues seen in this first controlled study of the JAK kinase inhibitor VVN461 ophthalmic solution in postoperative inflammation is promising and appropriate for further evaluation. By definition of its presumed mechanism of action, there would not be risk of corticosteroid-related AEs.

Acknowledgments

The VVN461-CS201 Investigators are listed in Table S1. The authors acknowledge Ben Ashberg, Lexitas Pharma Services, Inc for medical writing.

Footnotes and Disclosures

Originally received: January 28, 2025.
Final revision: April 15, 2025.
Accepted: April 18, 2025.
Available online: April 25, 2025. Manuscript no. XOPS-D-25-00061.

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This work is accepted as a presentation at the Association for Research in Vision and Ophthalmology annual meeting May 4-8, 2025, Salt Lake City, UT.

Disclosure(s):

All authors have completed and submitted the ICMJE disclosures form.

The author(s) have made the following disclosure(s):

J.B.: Grants – VivaVision Biotech, Inc.

K.Y.J.: Grants – VivaVision Biotech, Inc.

X.-Y.L.: Stock – VivaVision Biotech, Inc.; Employee – VivaVision Biotech, Inc.

C.L.: Stock – VivaVision Biotech, Inc.; Employee – VivaVision Biotech, Inc.

G.D.N.: Consultant – VivaVision Biotech, Inc.

J.H.P.: Grants – VivaVision Biotech, Inc.

W.S.: Stock – VivaVision Biotech, Inc.; Employee – VivaVision Biotech, Inc.

This study was supported by VivaVision Biotech, Inc.

Support for Open Access publication was provided by VivaVision Biotech, Inc.

HUMAN SUBJECTS: Human subjects were included in this study. The study was approved by an institutional review board, adhered to the Declaration of Helsinki, and all subjects provided written informed consent before enrollment in the study. This study was registered on clinicaltrials.gov as NCT06164743.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Li, Shen, Lu, Novack

Data collection: Peace, Jong, Bacharach

Analysis and interpretation: Peace, Jong, Bacharach, Li, Shen, Lu, Novack

Obtained funding: N/A

Overall responsibility: Peace, Jong, Bacharach, Li, Shen, Lu, Novack

Abbreviations and Acronyms:
ACC = anterior chamber cell; **AE** = adverse event; **JAK** = Janus kinase; **IOP** = intraocular pressure; **IP** = investigational product; **TEAE** = treatment emergent adverse event.

Keywords:
VVN461, Inflammation, Cataract surgery, JAK kinase.

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